R13

Code No: 111AD

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, May/June - 2019
ENGINEERING PHYSICS
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, ETM, MMT, AE, AME, MIE, PTM, AGE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	What is Ionic Bond?	[2]
b)	What is Hydrogen Bond?	[3]
c)	Write two important properties of Matter Waves.	[2]
d)	What is the main difference between Micro Canonical and Canonical Ensembles?	[3]
e)	What is Polarizability?	[2]
f)	What is the Magnetization?	[3]
g)	Write applications of lasers.	[2]
h)	Define Numerical Aperture.	[3]
i)	Explain about top down approach for synthesis of nanomaterials.	[2]
j)	What is Sabine's Formula? Explain.	[3]

PART-B

(50 Marks)

2.a)	What is packing fraction? Deduce packing factor for BCC structure.	
b)	Write notes on Frenkel and Schottky defects with diagrams.	[5+5]
	OR	
3.a)	Explain with suitable examples the covalent and metallic type of bondings in solids.	
b)	Write a note with neat diagram of Structure of Diamond.	[5+5]
4.a)	Derive an expression for Schrödinger's Time Independent Wave Equation.	
b)	Describe Concept of Effective Mass of an Electron.	[5+5]
	OR	
5.a)	Write an essay on Davisson and Germer's Experiment.	
b)	Explain Origin of Energy Band Formation in Solids.	[5+5]

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6.a) Derive an expression for electronic polarizability in terms of the radius of the atom.
b) Write short notes on piezoelectricity. [5+5]

OR

7.a) Write notes on Type-I and Type-II Superconductors along with the applications of Superconductors.
b) Discuss about Meissner Effect. [5+5]

8.a) Discuss about Diffraction grating experiment with the help of diagram.
b) Write short notes on Nicol prism. [5+5]

OR

9.a) Write an essay on construction and working of Helium-Neon Laser.
b) What is the importance of the Attenuation in Optical Fibers? [5+5]

10.a) Write an essay on Chemical Vapour Deposition.
b) Discuss about Solar cell, LED and Photo Diodes. [5+5]

OR

11.a) What are the factors affecting the architectural acoustics and suggest their remedies.
b) Write notes on Quantum Confinement. [5+5]

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